3.7.1 Instantaneous particle speed and kinetic energy

For simple harmonic particle motion in a wave, speed is greatest at equilibrium and momentarily zero at maximum displacement before reversal. Since $K = \frac{1}{2}mu^2$, kinetic energy is greatest in the first case and zero in the second. Zero displacement therefore does not mean rest. Particle speed varies during oscillation and differs from the waveform's propagation speed.

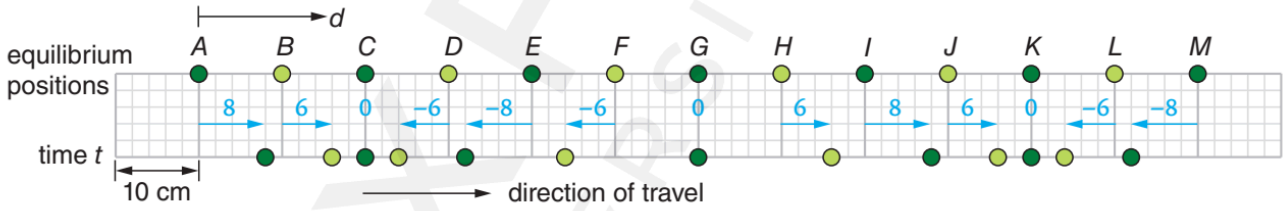


Fig 7.1g *A longitudinal wave at time t.*

Actual positions of particles A–M at the same instant. Blue arrows show horizontal displacement: right is positive and left is negative.

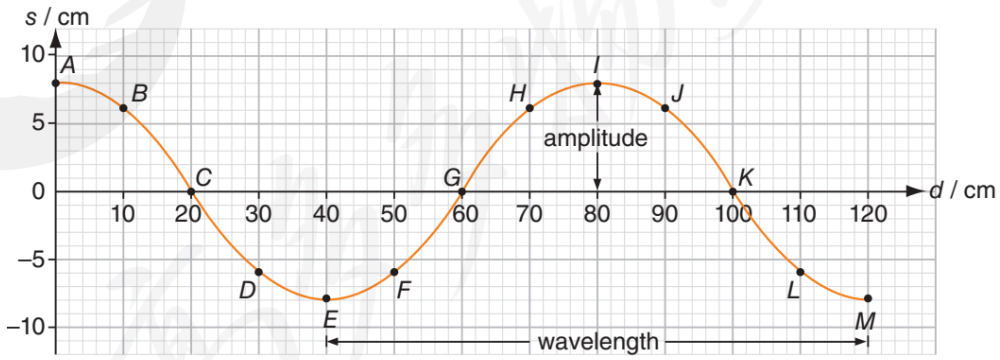
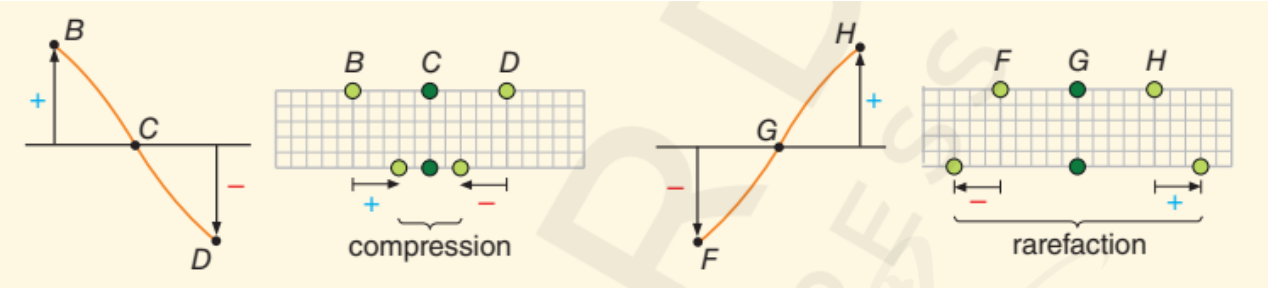


Fig 7.1h *The s–d graph of a longitudinal wave.*

Match A–M to the particles above. The horizontal axis gives equilibrium position; the vertical axis gives horizontal displacement. C, G and K have zero displacement and maximum speed; A, E, I and M have maximum displacement magnitude and are momentarily at rest. Graph height represents displacement; particles still oscillate horizontally.

3.7.2 Compression and rarefaction on displacement graphs

First establish that displacement to the right is positive. On a displacement-distance graph, a downward crossing of zero marks a compression centre. An upward crossing marks a rarefaction centre. Graph height represents displacement, not vertical particle motion. The neighbouring-particle diagram shows particles crowding towards a compression or separating away from a rarefaction.



Neighbouring particles crowd together in a compression and move apart in a rarefaction.

3.7.3 Sound speed and the medium

Sound requires a medium. Its speed depends on properties including elasticity, density and temperature. Common materials often transmit sound faster in solids than liquids and faster in liquids than gases, but this is not universal. With medium conditions fixed, increasing loudness or changing source frequency does not normally change sound speed directly.

$$v_{\text{solid}} > v_{\text{liquid}} > v_{\text{gas}}$$

Typical sound-speed ordering for common materials, not a universal inequality for every solid, liquid and gas. v: sound speed (m s^{-1}).

3.7.4 Reflection, refraction and interference of sound

Sound reflects, refracts, diffracts and interferes like other waves. Echoes result from reflection. Regions of different sound speed can bend propagation. Spreading around obstacles involves diffraction. Coherent sources can produce alternating loud and quiet positions.

3.7.5 Audible sound, infrasound and ultrasound

A young person with normal hearing typically hears about 20 Hz to 20 kHz. Lower frequencies are infrasound and higher frequencies are ultrasound. Actual limits vary with age and individual hearing. This classification concerns frequency, not loudness. Ultrasound is a mechanical wave, not electromagnetic radiation, and cannot travel through a vacuum.

$$20\ \text{Hz} \lessapprox f \lessapprox 20\,000\ \text{Hz}$$
$$f > 20\,000\ \text{Hz}$$

Approximate audible range for a young healthy human; sensitivity and upper limit vary with age and the individual.

Ultrasound has frequency above approximately 20 kHz, the conventional upper limit of human hearing.

3.7.6 Applications of ultrasound

Transmitted ultrasound and returning echoes locate fish, scan structures and reveal cracks in materials. Other textbook applications include cleaning delicate objects and concentrating energy to break kidney stones.

3.7.7 Pitch, loudness and quality

Pitch generally rises with frequency. At comparable frequency and conditions, greater amplitude usually sounds louder. Quality or timbre depends on waveform and harmonic content, allowing instruments with the same pitch and loudness to sound different.

3.7.8 Typical sound intensity levels

Textbook examples include 20 dB in a country park, 40 dB for soft conversation, 60 dB for ordinary conversation and 70 dB for a vacuum cleaner 3 m away. Zero dB is a hearing-threshold reference, not the absence of sound. Use any specified distances and measurement conditions when comparing examples.